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FlowMerce

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Graduation Project

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Midyear Short Documentation

Project Short Documentation

The document includes the following content:

1. Abstract:

The rapid growth of e-commerce has created significant opportunities for businesses to expand their market reach; however, many small and medium-sized enterprises (SMEs) continue to face challenges in building competitive and high-performing online stores. These challenges go beyond product availability and pricing, and often include weak customer experience, ineffective storefront design, limited data-driven decision-making, and high costs associated with professional design and business analysis services.

Flowmerce proposes a smart e-commerce platform designed to support merchants in managing and growing their online businesses more effectively. The platform provides an integrated environment that enables merchants to create and operate digital storefronts, manage products and orders, and gain valuable insights into sales and customer behavior. A key differentiator of the project is the integration of an AI-powered assistant that acts as a business and UI/UX advisor, offering strategic recommendations to improve store appearance, customer engagement, and conversion performance.

By combining intelligent guidance with comprehensive business management capabilities, Flowmerce aims to empower SMEs to compete in the digital marketplace, enhance operational efficiency, and achieve sustainable growth.

2. Background

a. Introduction to the Main Idea

With the rapid growth of digital commerce, online shopping platforms have become essential tools for businesses of all sizes. However, many merchants, especially small and medium enterprises, struggle to design, manage, and optimize professional online storefronts without advanced technical knowledge. Traditional e-commerce platforms often provide basic tools for product management and order processing but lack intelligent guidance to help merchants improve user experience and conversion rates.

This project proposes an AI-assisted e-commerce platform that enables merchants to create, manage, and customize online stores while offering customers a smooth and secure shopping experience. The platform integrates artificial intelligence as a UX/UI advisor to analyze storefront designs and provide smart recommendations that enhance usability and accessibility.

b. Motivation

The motivation behind this project stems from several key challenges observed in existing e-commerce systems:

- Many merchants lack design expertise, leading to poorly optimized storefronts that reduce customer engagement and sales.
- Current platforms focus mainly on transactions, with limited support for intelligent decision-making and design optimization.

- Customers often abandon shopping carts due to complex checkout flows or unclear interfaces.
- Small businesses face difficulties competing with large enterprises that have access to professional designers and analysts.

By integrating AI-driven recommendations, analytics, and automation, this project aims to **bridge the gap between technical complexity and business usability**, empowering merchants to improve their stores without requiring deep technical or design knowledge.

c. Beneficiaries

The proposed system benefits multiple stakeholders:

- **Merchants:**
Gain access to tools for managing products, orders, analytics, and storefront customization, supported by AI recommendations that improve design and conversion rates.
- **Customers (Buyers):**
Enjoy a smooth shopping experience with intuitive navigation, secure checkout, multiple payment options, and order tracking.
- **Platform Administrators:**
Can efficiently manage merchants, monitor transactions, moderate content, and ensure platform security and compliance.
- **Delivery Service Providers (DSP):**
Benefit from clear order tracking, inventory synchronization, and automated status updates.
- **The Business Ecosystem:**
Supports digital transformation for small and medium businesses and encourages innovation in online commerce.

d. Main Techniques

The system relies on several modern technologies and techniques, including:

- **Web-based architecture** for accessibility and scalability.
- **Role-Based Access Control (RBAC)** to manage permissions for users, merchants, admins, and system services.
- **JWT-based authentication** to ensure secure user sessions.
- **AI and Machine Learning techniques** for analyzing storefront layouts and providing UX/UI recommendations.
- **API-based integration** with external services such as payment gateways, messaging services, and AI models (e.g., OpenAI/DeepSeek).
- **Data analytics** for tracking sales, customer behavior, and conversion performance.
- **Secure communication protocols (HTTPS/TLS)** to protect data in transit.

e. Main Application

The main application of this project is a **smart e-commerce platform** that allows merchants to launch and manage online stores with minimal effort while maximizing performance and usability. The system supports:

- Online store creation and publishing
- Product catalog management
- Shopping cart and checkout processing
- Secure payment handling
- Order tracking and delivery coordination

- AI-assisted storefront optimization
- Analytics and reports for business growth

This platform can be applied in various domains such as retail, fashion, electronics, local businesses, and service-based online stores, making it a flexible and scalable solution for modern digital commerce.

3. Problem definition

- Despite the widespread adoption of e-commerce platforms, many small and medium-sized merchants face challenges in developing effective and professional online stores. Existing platforms often require technical expertise and design knowledge to deliver high-quality user experience. This limitation results in poorly structured storefronts that negatively impact usability, customer engagement, and sales performance.
- **Reasons Behind This Problem**
 - i. Limited technical and UX/UI design knowledge among merchants.
 - ii. High complexity of current e-commerce platforms.
 - iii. Lack of intelligent guidance for improving storefront usability and design quality.
 - iv. High cost of professional design and development services.
 - v. Increasing user expectations for usability and accessibility.

4. Related work:

- Existing e-commerce platforms such as **Shopify**, **Salla**, **WooCommerce**, and **BigCommerce** provide core functionalities including store creation, product management, order processing, payment integration, and basic analytics.
- These platforms primarily focus on transactional operations and offer limited support for intelligent UX/UI evaluation or data-driven storefront optimization.
- Store customization in current solutions is largely template-based and often requires external plugins, professional designers, or manual interpretation of analytics, which increases cost and complexity for small and medium-sized enterprises.
- While some platforms incorporate AI-driven features, these are typically limited to backend analytics or product recommendations and do not provide actionable guidance for improving storefront usability and conversion rates.
- **Flowmerce** distinguishes itself by integrating an AI-powered assistant that analyzes storefront design and user experience, offering explainable and optional recommendations that merchants can accept or reject.
- The platform follows a modular architecture where payment gateways, delivery services, and AI models are treated as external services, ensuring flexibility, scalability, and maintainability.
- By combining standard e-commerce functionality with intelligent UX/UI guidance, Flowmerce addresses a gap in existing platforms and provides a more accessible and effective solution for SMEs seeking sustainable growth.

5. Project specifications

- System architecture

1- Architecture Style

FlowMerce is designed using a **microservices-based architecture** to ensure security isolation, scalability, and system reliability. The system is composed of independently deployable services, each responsible for a specific business capability, and communicating through secure RESTful APIs. The system follows a layered microservices architecture, where each service owns its business logic and data, ensuring loose coupling and high cohesion.

2- API Gateway

An **API Gateway** is used as a single entry point for client requests, handling authentication, request routing, and rate limiting

Role: JWT validation, Routing, Security isolation, Hide internal services

3- Authentication & Security Layer

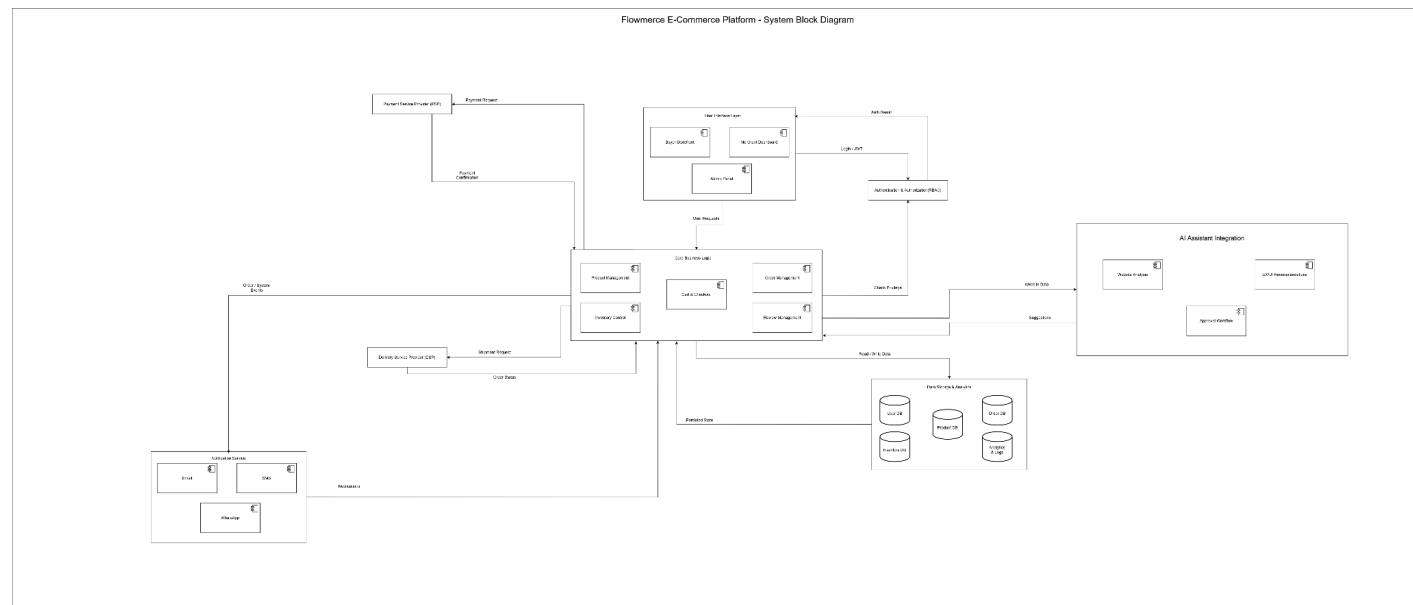
Authentication and authorization are handled using JWT-based security with role-based access control (RBAC)

4- Database Strategy

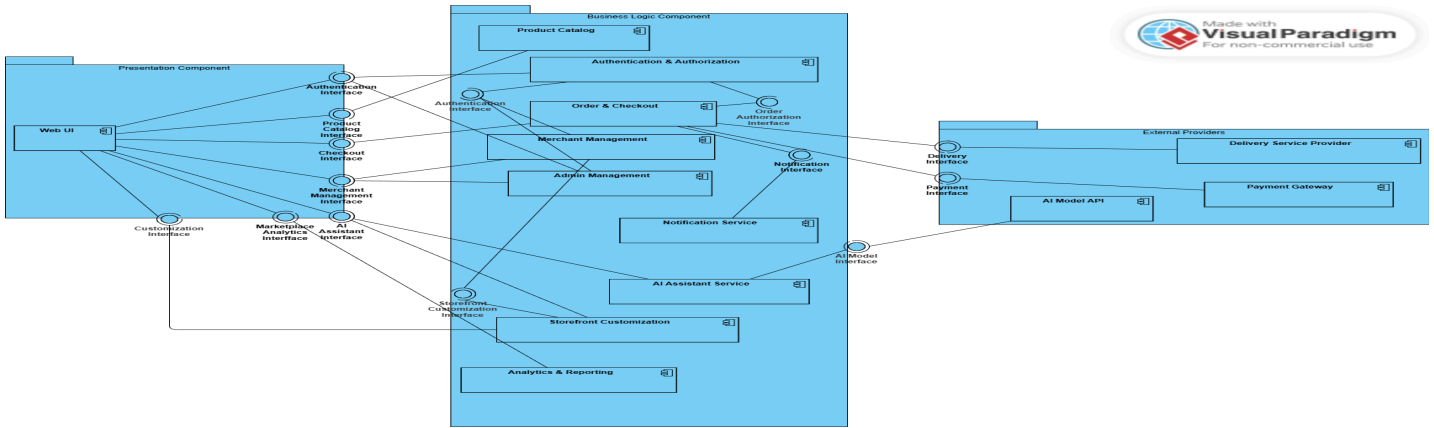
Each microservice owns its database schema to prevent tight coupling and ensure data isolation

5- Inter-Service Communication

Synchronous communication is handled via RESTful APIs, while asynchronous communication is achieved using RabbitMQ for events such as order placement, payment confirmation, and notifications



- Component diagram



● Stakeholders:

❖ Merchants

❖ Customers/Buyers

❖ System Administrators

❖ AI Assistant

❖ Payment Service Providers

❖ Delivery Service Providers

❖ Development Team

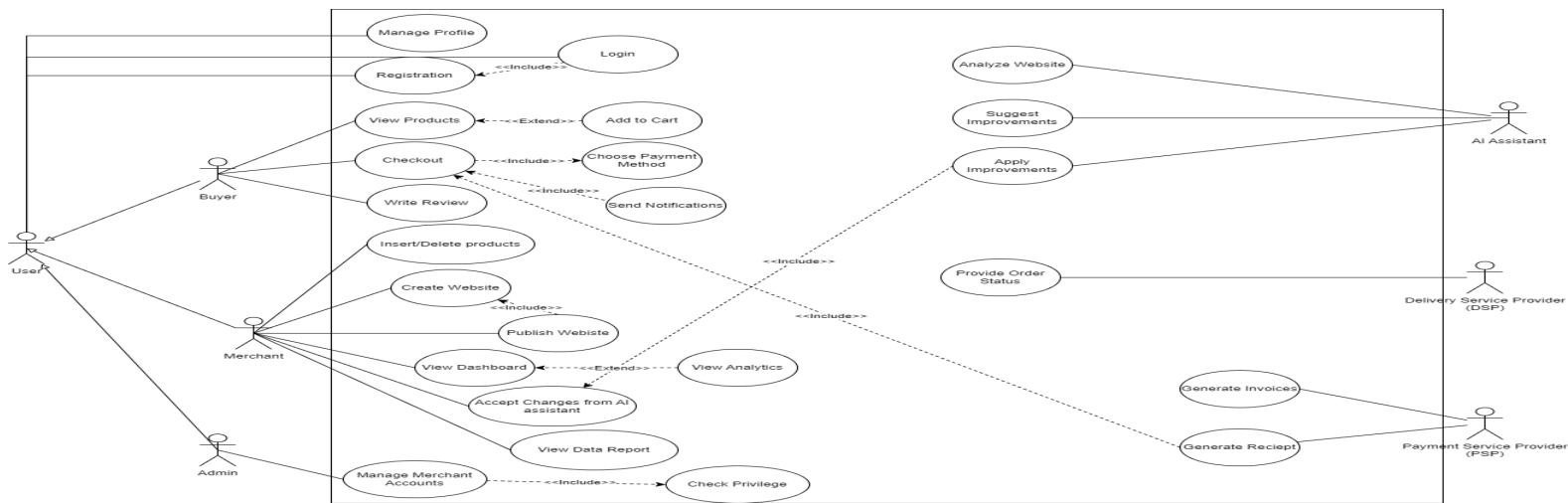
● Functional Requirements

Requirement	Description	Key Functions
1. User Management	Manages user-related operations including registration, authentication, authorization, and profile management.	● User registration with email and password ● Role-based access control (Merchant, Customer, Guest, Admin) ● Login/Logout and session management (using JWT) ● Password reset and profile management ● Multi-factor authentication (optional)
2. Merchant Dashboard	Provides merchants with tools to manage their stores, products, orders, and business insights.	● Create and manage multiple stores per account ● Full product lifecycle management (CRUD: name, price, inventory, variants, media) ● Order management: view, update, and fulfill orders ● Sales analytics and performance dashboards (lower priority) ● Manage coupons, discounts, and promotional campaigns (optional) ● AI-powered recommendations for storefront UI/UX improvements
3. Product Catalog & Search	Enables customers to browse, search, and explore products with efficient navigation and detailed	● Product listing with filtering and sorting (price, category, rating) ● Product detail view (media, description, variants) ● Smart search functionality (AI-assisted search ranking, optional) ● Customer product reviews and ratings system

	product information.	
4. Shopping Cart & Checkout	Provides customers with a seamless purchasing experience, from adding products to finalizing orders.	● Add, remove, and update items in the cart ● Persistent cart per session or registered user ● Guided checkout flow (address, shipping, payment selection) ● Integration with multiple payment gateways (Stripe, Mada, STC Pay) ● Order confirmation and invoice generation ● Abandoned cart notifications (email/SMS)
5. Storefront Customization	Enables merchants to design and personalize their storefronts to match their brand identity and improve customer experience.	● Theme selection and customization options ● Drag-and-drop page builder for layouts (to be discussed) ● Manage navigation menus, banners, and color schemes ● AI Assistant recommendations (e.g., color contrast, layout balance, accessibility) ● Preview and publish store changes
6. AI Assistant (UX/UI Advisor)	Acts as a core differentiator by providing intelligent design and UX guidance to help merchants optimize their storefronts.	● Analyze current store layout and design ● Suggest improvements (color palette, visual hierarchy, typography) ● Provide accessibility feedback (contrast ratios, readability) ● Offer layout templates optimized for conversion rates ● Integrate with OpenAI/DeepSeek API for natural-language design advice
7. Order Management	Handles order lifecycle management, tracking, and fulfillment while keeping customers and merchants updated.	● Track order status (Pending, Shipped, Delivered, Canceled) ● Generate invoices and receipts ● Process returns and refunds ● Send notifications via email, SMS, or WhatsApp ● Automatically adjust inventory upon order completion
8. Analytics & Reports	Provides merchants with data-driven insights to track performance, understand customer behavior, and optimize business outcomes.	● Sales analytics (daily, weekly, monthly trends) ● Customer segmentation and top products reporting ● Conversion tracking (visitors → sales) ● AI-driven insight suggestions (e.g., “Customers abandon at checkout — simplify process”)
9. Communication & Notification	Ensures merchants and customers stay informed through timely updates and alerts across multiple channels.	● Email and SMS notifications for order status and promotions (opt) ● System notifications within the merchant dashboard ● Integration with third-party messaging services (WhatsApp Business, Email)
10. Security & Compliance	Safeguards platform data, ensures user privacy, and enforces regulatory compliance.	● Data encryption at rest and in transit (HTTPS/TLS) ● Role-based permissions and access control ● Compliance with GDPR and local regulations (e.g., Saudi e-commerce law)

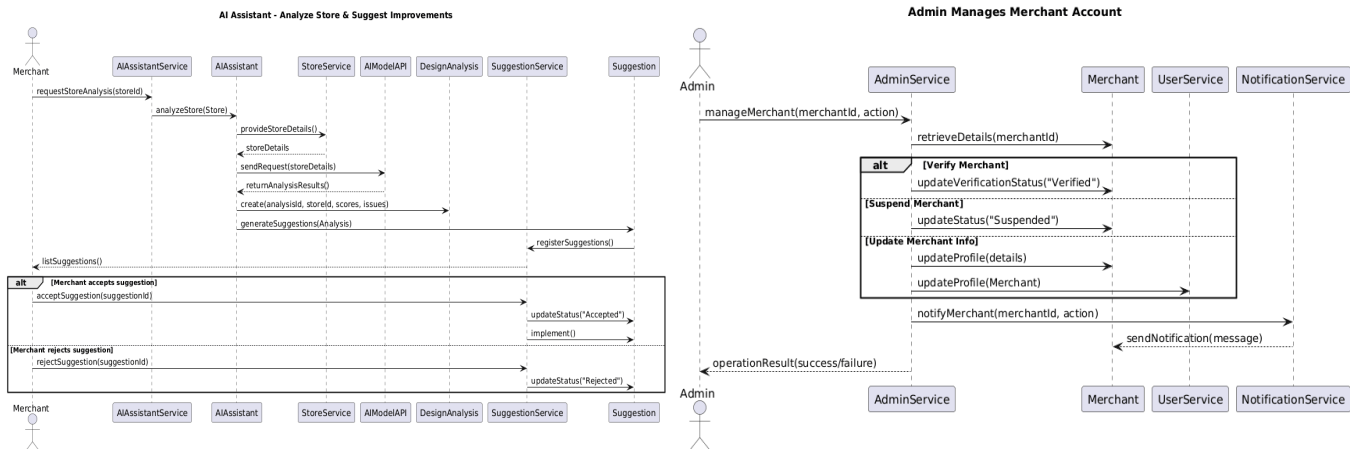
		• Audit logs for administrator actions
11. Admin Panel	Serves as the internal control center for platform administrators to manage users, monitor activity, and configure system settings.	• Manage merchant accounts • Monitor transactions and detect fraud • Moderate content (reviews, comments) • Configure system settings, feature toggles, and perform backups

- Non-functional Requirements
 - 1. Security Requirements**
 - User authentication and authorization using secure mechanisms.
 - Encryption of sensitive data during storage and transmission.
 - Secure payment processing through trusted third-party providers.
 - Protection against common web vulnerabilities (e.g., SQL injection).
 - Role-based access control for merchants, administrators, and system components.
 - 2. Performance Requirements**
 - The system shall respond to user requests within acceptable time limits under normal load.
 - The platform shall support concurrent users without performance degradation.
 - 3. Scalability Requirements**
 - The system shall scale horizontally to support growth in merchants and customers.
 - Cloud-based deployment to handle increased traffic and data volume.
 - 4. Reliability & Availability**
 - The system shall ensure high availability with minimal downtime.
 - Regular backups and recovery mechanisms shall be implemented.
 - 5. Usability Requirements**
 - The platform shall provide an intuitive and user-friendly interface.
 - AI recommendations shall be easy to understand and actionable.
 - The system shall support responsive design for multiple devices.
 - 6. Accessibility Requirements**
 - The system shall comply with basic accessibility standards.
 - Support for readable typography, color contrast, and keyboard navigation.
 - 7. Maintainability**
 - The system shall be modular to allow easy updates and enhancements.
 - Clear documentation for future development and maintenance.
- Use-case Diagram



• Class Diagram

• Sequence Diagram



6- Preferred Payment Methods : Most common is Cash on Delivery

7- Pricing Model Preferences : Users favor either a free model with limited features

8- Most Desired Features : Easy drag-and-drop websitebuilder, AI-powered design suggestions, Advanced analytics and reports

9- Additional Features : Maybe it could come with a hosting package with it? like some people think it's a hassle to make a site then host it on a different platform so maybe the ease of that would encourage them to do it more

Enhancing in finding bugs and fixing code errors that are created by it

Tell AI your budget and goals, it builds the perfect store plan for you with best possible options

Task ID	Task Title	Description	Dependencies	Suggested Phase / Order	Status
T1	Conduct problem research & market survey	Study SME e-commerce challenges, analyze existing platforms (Shopify, Salla, WooCommerce, BigCommerce), and identify gaps related to UX/UI optimization and intelligent guidance.	–	Analysis – very early	Completed
T2	Define project vision, scope, beneficiaries & personas	Define Flowmerce goals, scope, target SMEs, stakeholders (merchants, customers, admins, DSPs, AI assistant), and user personas.	T1	Analysis – early	Completed
T3	Draft abstract & background sections	Write abstract, introduction, motivation, beneficiaries, main techniques, and main application for the documentation.	T1, T2	Documentation – early	Completed
T4	Document functional requirements & actors	Define functional requirements and actors for all core modules (user management, catalog, storefront, orders, AI assistant, analytics, notifications, admin).	T2	Analysis – early	Completed
T5	Document non-functional requirements	Specify security, performance, scalability, usability, accessibility, reliability, and maintainability requirements.	T2, T4	Analysis – early	Completed
T6	Identify stakeholders & main techniques	Finalize stakeholder list and core technologies (web stack, RBAC, JWT, AI APIs, analytics, HTTPS/TLS).	T2, T4	Analysis – early	Completed
T7	Design system architecture (block diagram)	Design high-level system architecture showing major services, data flows, and external integrations (payments, DSPs, AI providers).	T4, T5, T6	Architecture – early	Completed
T8	Design component diagram	Refine architecture into well-defined components with clear responsibilities and interfaces, aligned with UML 2.0 standards.	T7	Architecture – mid	Completed

T9	Design and Choose Branding Artifacts	Define the visual identity of the Flowmerce platform by selecting and documenting branding artifacts, including color palette, typography (fonts), logo concept, iconography style, and overall design guidelines.	T7, T8	Design – mid (UI/UX & branding foundation)	Completed
T10	Write use-case scenarios & refine use-case diagram	Define detailed use-case scenarios and refine the UML use-case diagram with main and alternative flows.	T4, T6	Analysis / Design – current	Completed
T11	Design database ERD	Design the entity–relationship diagram covering users, stores, products, orders, payments, shipments, analytics events, and notifications.	T4, T6, T10	Design – mid	Completed
T12	Design abstract class diagram	Develop abstract class diagram by package, focusing on responsibilities, relationships, and design patterns without low-level implementation details.	T4, T6, T10, T11	Design – mid	Completed
T13	Design sequence diagrams	Create sequence diagrams for key system flows (authentication, browsing, checkout, payment, AI recommendation, order lifecycle).	T4, T6, T10	Design – mid / late	Completed
T14	Define technology stack & learning plan	Finalize frameworks, languages, databases, APIs, and define learning tasks required by team members.	T1, T2, T8, T12,	Planning – parallel	Completed
T15	Build detailed work plan & Gantt chart	Convert tasks into a structured work plan and Gantt chart to manage execution and milestones.	T1–T8, T13	Planning – current	Completed

T16	Design UI/UX Wireframes and Interactive Prototypes (Figma)	Design low-fidelity and high-fidelity UI/UX wireframes on Figma for core Flowmerce screens (merchant dashboard, storefront, cart/checkout, admin, AI assistant), ensuring consistency with branding artifacts and system flows.	T8.5, T10, T13	Design – late (UI/UX design & validation)	Completed
T17	Conduct UI/UX Usability Evaluation and Refinement	Evaluate the designed UI/UX wireframes using basic usability testing and heuristic evaluation for key user flows, collect feedback, identify usability issues, and refine designs accordingly.	T16	Design – late (usability validation & refinement)	Completed
T18	Supervisor feedback & validation meeting	Review analysis and design artifacts with supervisor and confirm readiness for the final discussion	T3–T17	Planning – after design	Completed
T19	Implement core platform foundation	Set up repository, authentication, RBAC, user management, and baseline security mechanisms.	T4, T5, T7, T8, T13	Implementation – start	In-Progress
T20	Implement merchant & store management	Develop merchant dashboard, store setup, product and order management, and inventory synchronization.	T16	Implementation – early / mid	Planned
T21	Implement customer journey & payments	Implement catalog browsing, cart, checkout flow, payment gateway integration, and order confirmation.	T16, T17	Implementation – mid	Planned
T22	Implement storefront customization & AI assistant MVP	Develop theme-based storefront customization and integrate AI assistant for optional UX/UI recommendations.	T16, T18	Implementation – mid / late	Planned
T23	Implement analytics, notifications & DSP integration	Integrate analytics, sales reports, asynchronous notifications (email/SMS), and delivery service providers.	T16–T19	Implementation – late	Planned
T24	Testing & validation	Define and execute unit, integration, usability, and basic performance testing	T16–T20	Testing – late	Planned

		for core flows.			
T25	Deployment & operations setup	Prepare CI/CD pipeline, staging/production environments, monitoring, logging, and backup strategies.	T16–T20	Deployment – late	Planned
T26	Maintain and refine Final Year documentation	Ensure consistency between documentation, diagrams, and agreed design decisions.	T3–T12	Documentation – current	In progress
T27	Final report, presentation & demo	Prepare final documentation, presentation slides, and system demo for project evaluation.	T16–T22, T23	Documentation – final	Planned